Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 02: Database and Basic SQL

Digital for Executive

Outline

- What is Database?
- Why we need Database?
- Relational Model Constraint
- SELECT TABLE Statement
- SELECT DISTINCT
- LIMIT
- COUNT
- AVG
- SUM
- WHERE
- ORDER BY

What is Database?

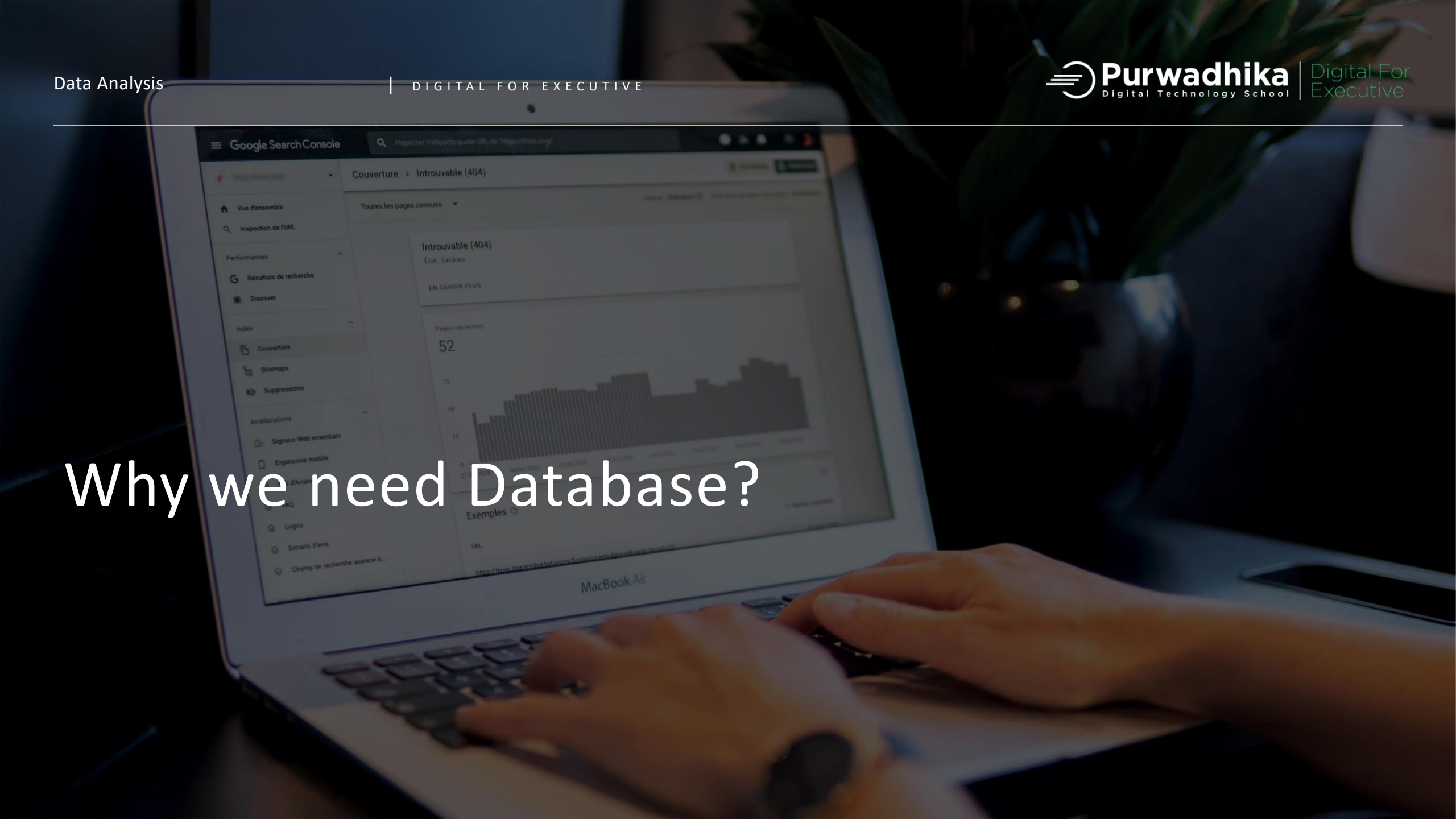
Database Definition

- A **database** is an organized collection of data, generally stored and accessed electronically from a computer system.
- **Database**, also called electronic database, any collection of data, or information, that is specially organized for rapid search and retrieval by a computer.
- Where databases are more complex they are often developed using formal design and modeling techniques.

Database Terminology

- Formally, a "database" refers to a set of related data and the way it is organized.
- Access to this data is usually provided by a "database management system" (DBMS) consisting of an integrated set of computer software that allows users to interact with one or more databases and provides access to all of the data contained in the database (although restrictions may exist that limit access to particular data).

Why we need Database?



Database Functionality

1. Database Management System (DBMS) help to correlate, query, and report to the collected information.
2. DBMS can quickly answer lots of questions.
3. DBMS help to understand complicated data.
4. DBMS can improve our work and reducing excessive work hours in pursuit of particular objectives.

Using Database to Improve Business Performance and Decision-Making

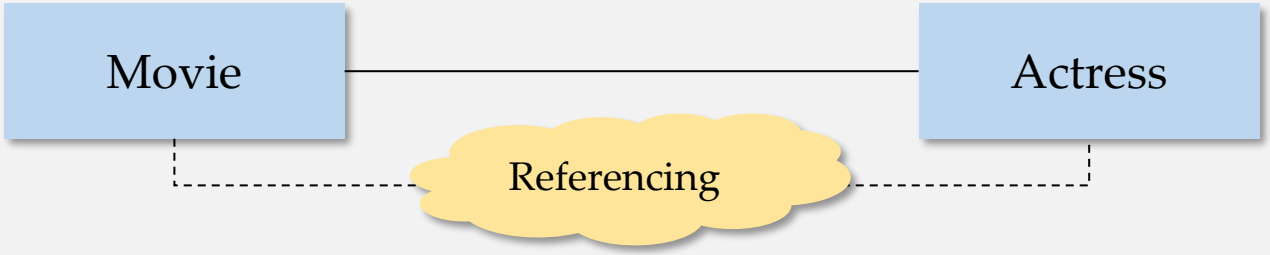
Businesses use their databases to:

1. Keep track of basic transaction
2. Provide information that will help the company run the business more efficiently
3. Help managers and employees make better decisions

Relational Model Constraint

Referencing

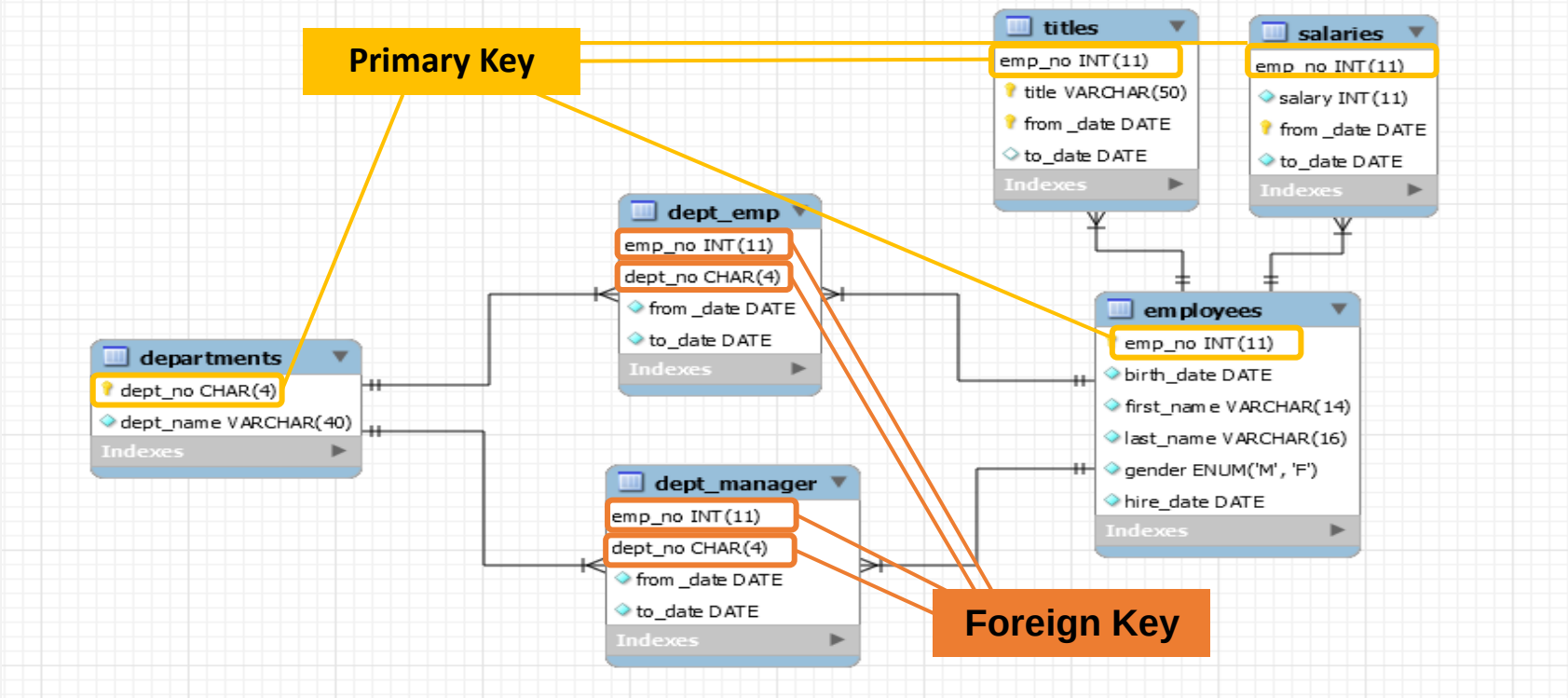
- In any business, data must adhere to certain restrictions or rules.
- For example, relationship between actress and movie.
- At least, **one actress** to be actor in **one movie**. This is a **one to one** relationship.
- To look up the actress information, the movie entity refers to the actress entity.
  To look up the movie information, the actress entity refers to the movie entity.
  In a relational data model this is called **referencing.**



Primary Key & Foreign Key

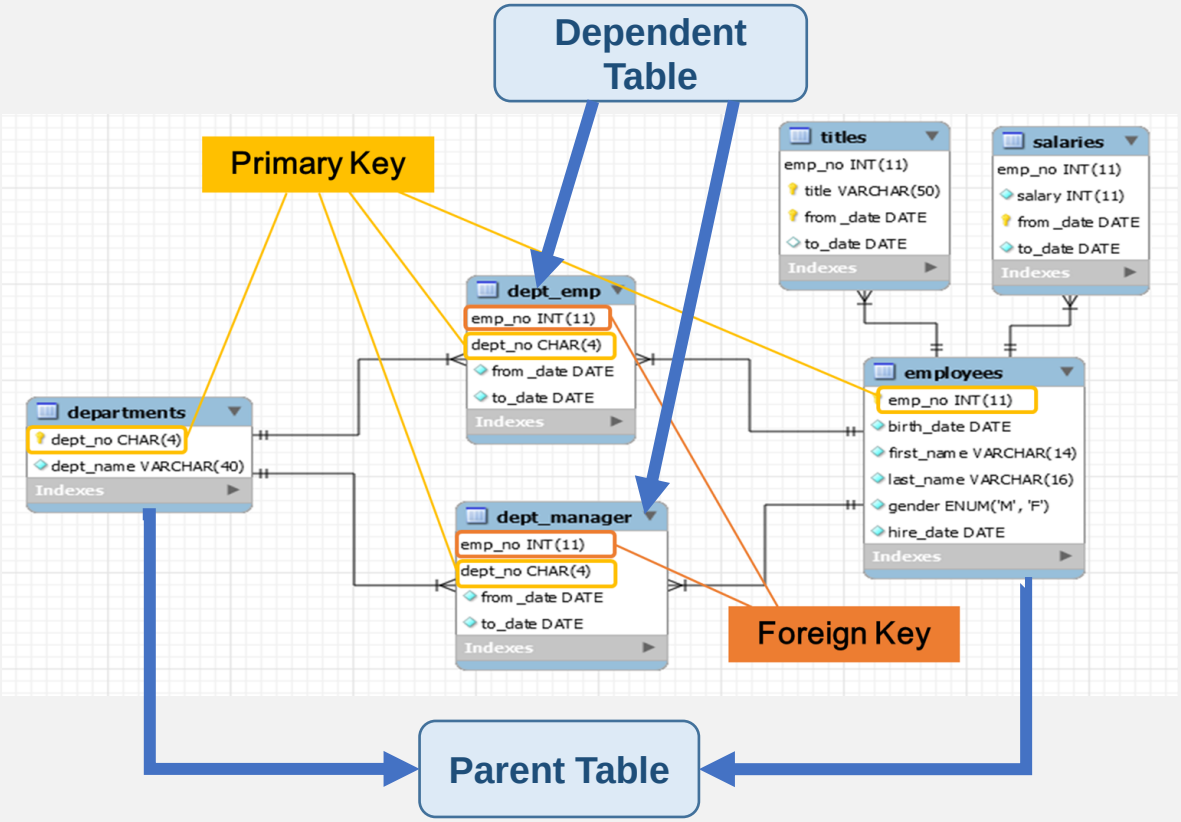
- Every table have **unique identity**, such as salary_id, employee_id, department_id, etc.This identifies the primary key
- The primary key constraint uniquely identifies each record in a table.
- Primary keys must contain UNIQUE values, and cannot contain NULL values.
- But, sometimes **“A” table primary key** is placed in **“B” table**. This ‘primary key’ is called **foreign key** in “B” table.
- A foreign key is a set of columns referring to a primary key of another table. A foreign key is a key **used to link two tables together**.
- A foreign key is a field (or collection of fields) in one table that refers to the primary key in another table.

ERD Representation of a Relational Data Model



Relational Model Terminology

- **Parent table:** a table containing a Primary Key that is related to at least one Foreign Key.
- **Dependent table:** a table containing one or more Foreign Keys

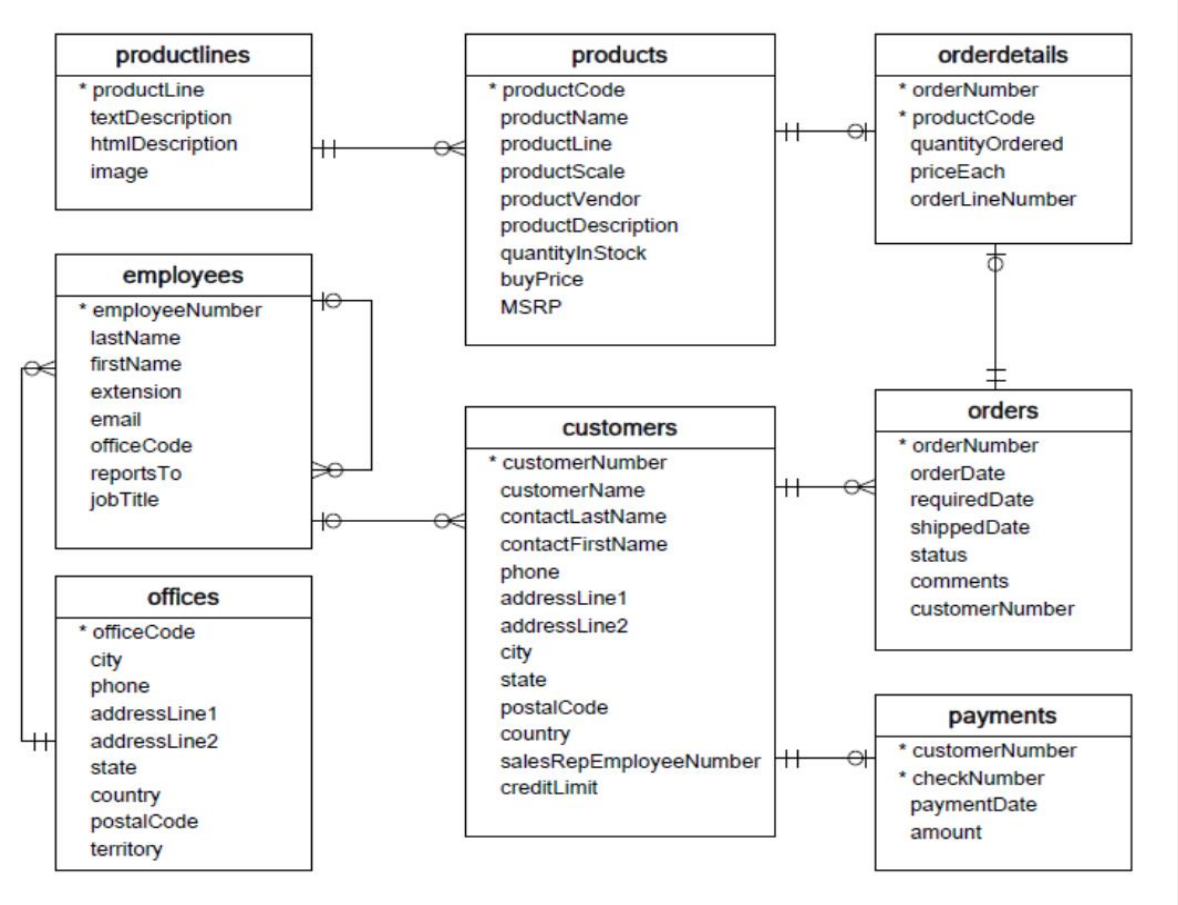


Create Classicmodels Database

1. Open “classicmodels database.sql” file using any text editor
2. In MySQL Workbench, Run the script contained in the file
3. Run syntax “use classicmodels;”

Database scheme can be seen on the left

*you will use this database for practice in the upcoming sessions

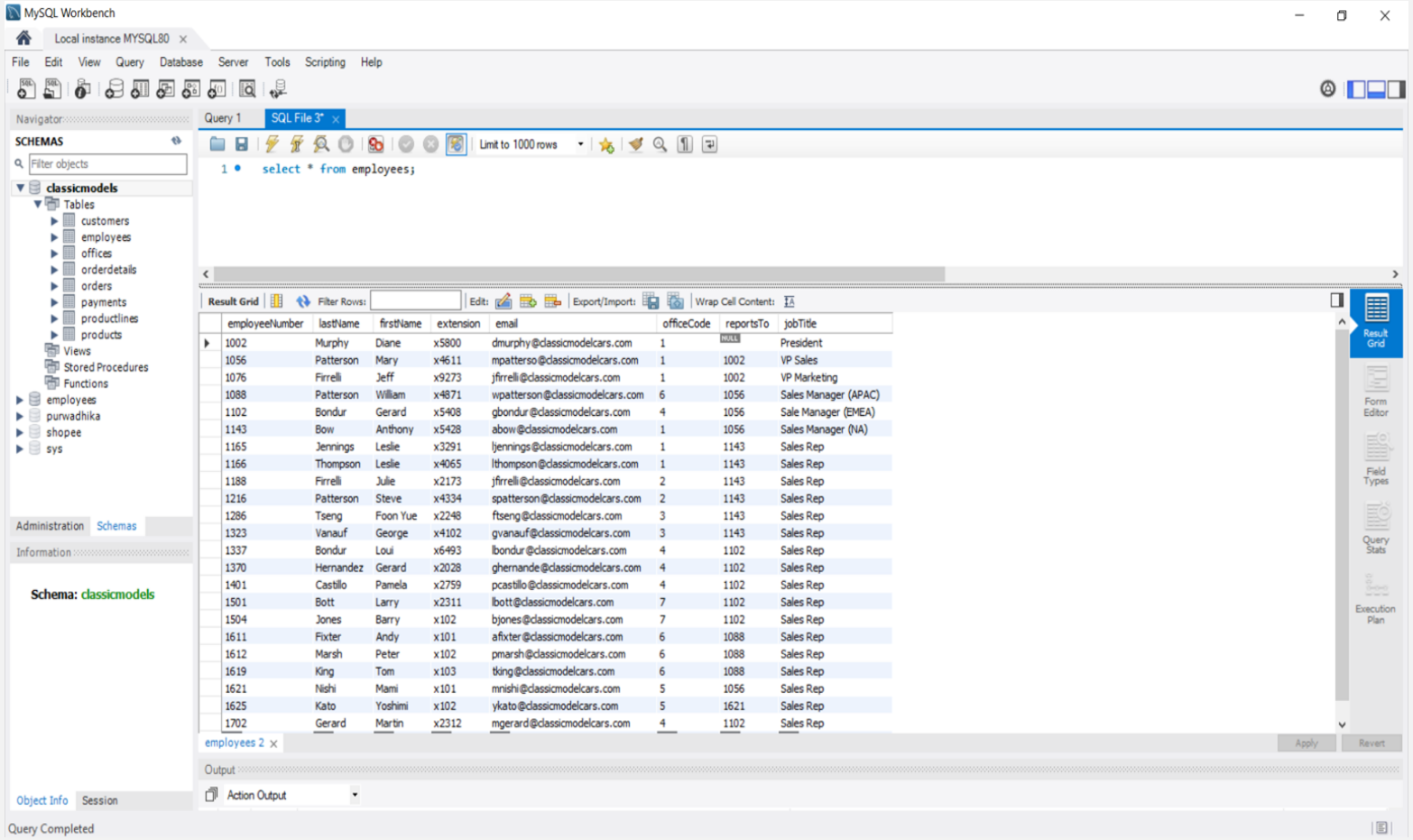


Source : https://www.mysqltutorial.org/mysql-sample-database.aspx/https://www.mysqltutorial.org/mysql-sample-database.aspx/

Create Classicmodels Database

You can see. This is the result if you successfully added the classicmodels database

Try query :
Select * from employees;



employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing
1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep
1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep
1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep
1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102	Sales Rep
1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1102	Sales Rep
1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1102	Sales Rep
1501	Bott	Larry	x2311	lbott@classicmodelcars.com	7	1102	Sales Rep
1504	Jones	Barry	x102	bjones@classicmodelcars.com	7	1102	Sales Rep
1611	Fixter	Andy	x101	afixter@classicmodelcars.com	6	1088	Sales Rep
1612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	6	1088	Sales Rep
1619	King	Tom	x103	tking@classicmodelcars.com	6	1088	Sales Rep
1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1056	Sales Rep
1625	Kato	Yoshimi	x102	ykato@classicmodelcars.com	5	1621	Sales Rep
1702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	4	1102	Sales Rep

SELECT TABLE Statement

SELECT TABLE Statement

- The SELECT statement is used to select data from a database.
- The data returned is stored in a result table, called the result-set.

```
SELECT column1, column2, ...
FROM table_name;
);
```

- Here, column1, column2, ... are the field names of the table you want to select data from. If you want to select all the fields available in the table, use the following syntax:

```
SELECT * FROM table_name;
```

Example

- The following example select all column in a table called “orders" that contains seven columns: orderNumber, orderDate, requiredDate, shippedDate, status, comments, customerNumber

```
SELECT * FROM orders;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
10100	2003-01-06	2003-01-13	2003-01-10	Shipped	NULL	363
10101	2003-01-09	2003-01-18	2003-01-11	Shipped	Check on availability.	128
10102	2003-01-10	2003-01-18	2003-01-14	Shipped	NULL	181
10103	2003-01-29	2003-02-07	2003-02-02	Shipped	NULL	121
10104	2003-01-31	2003-02-09	2003-02-01	Shipped	NULL	141
10105	2003-02-11	2003-02-21	2003-02-12	Shipped	NULL	145
10106	2003-02-17	2003-02-24	2003-02-21	Shipped	NULL	278
10107	2003-02-24	2003-03-03	2003-02-26	Shipped	Difficult to negotiate with customer. We need m...	131
10108	2003-03-03	2003-03-12	2003-03-08	Shipped	NULL	385
10109	2003-03-10	2003-03-19	2003-03-11	Shipped	Customer requested that FedEx Ground is used	486

Example

- Select only column orderNumber, status and customerNumber

```
SELECT orderNumber, status, customerNumber FROM orders;
```

```
1 • SELECT orderNumber, status, customerNumber FROM orders;
```

Result Grid | Filter Rows: | Edit: | Export/Imp

orderNumber	status	customerNumber
10100	Shipped	363
10101	Shipped	128
10102	Shipped	181
10103	Shipped	121
10104	Shipped	141
10105	Shipped	145
10106	Shipped	278
10107	Shipped	131
10108	Shipped	385
10109	Shipped	486

SELECT DISTINCT

SELECT DISTINCT

- The SELECT DISTINCT statement is used to return only distinct (different) values.
- Inside a table, a column often contains many duplicate values; and sometimes you only want to list the different (distinct) values.

```
SELECT DISTINCT column1, column2, ...
FROM table_name;
```

SELECT DISTINCT

- The following SQL statement selects only the DISTINCT values from the "**jobTitle**" column in the "employees" table:

```
SELECT DISTINCT jobTitle
FROM employees;
```

jobTitle
President
VP Sales
VP Marketing
Sales Manager (APAC)
Sale Manager (EMEA)
Sales Manager (NA)
Sales Rep

LIMIT

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LIMIT

- The following SQL statement selects the first **three** records from the “employees" table

```
SELECT * FROM employees LIMIT 3;
```

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing

LIMIT

- The following SQL statement selects the first **five** records from the “City" table

```
SELECT * FROM orders LIMIT 5;
```

orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
10100	2003-01-06	2003-01-13	2003-01-10	Shipped	NULL	363
10101	2003-01-09	2003-01-18	2003-01-11	Shipped	Check on availability.	128
10102	2003-01-10	2003-01-18	2003-01-14	Shipped	NULL	181
10103	2003-01-29	2003-02-07	2003-02-02	Shipped	NULL	121
10104	2003-01-31	2003-02-09	2003-02-01	Shipped	NULL	141

COUNT

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COUNT

- The COUNT() function returns the number of rows that matches a specified criterion.

```
SELECT COUNT(column_name)
FROM table_name;
```

COUNT

- The following SQL statement finds the number of **employeeNumber** from **employees** table:

```
SELECT COUNT(employeeNumber) FROM employees;
```

	COUNT(employeeNumber)
▶	23

COUNT

- The following SQL statement finds the number of **jobTitle** from employees table:

```
SELECT COUNT(distinct jobTitle) FROM employees;
```

	COUNT(distinct jobTitle)
▶	7

COUNT

- The following SQL statement finds the number of **jobTitle** from employees table and renames the column:

```
SELECT
    COUNT(DISTINCT jobTitle) AS number_of_jobtitle
FROM
    employees;
```

	number_of_jobtitle
▶	7

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AVG

AVG

- The AVG() function returns the average value of a numeric column.

```
SELECT AVG(column_name)
FROM table_name;
```

AVG

- The following SQL statement finds the average **amount** from **payments** table:

```
SELECT
    AVG(amount)
FROM
    payments;
```

	AVG(amount)
▶	32431.645531

AVG

- The following SQL statement finds the average **amount** from **payments** table and renames the column:

```
SELECT
    AVG(amount) as average_payments
FROM
    payments;
```

	average_payments
▶	32431.645531

SUM

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SUM

- The SUM() function returns the total sum of a numeric column.

```
SELECT SUM(column_name)
FROM table_name;
```

SUM

- The following SQL statement finds the total **items sold** from orderdetails table:

```
SELECT
   SUM(quantityOrdered)
FROM
   orderdetails;
```

	SUM(quantityOrdered)
▶	105516

SUM

- The following SQL statement finds the total **items sold** from orderdetails table and renames the column:

```
SELECT
   SUM(quantityOrdered) as items_sold_qty
FROM
   Orderdetails;
```

	items_sold_qty
▶	105516

WHERE

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WHERE

- The WHERE clause is used to filter records.
- The WHERE clause is used to extract only those records that fulfill a specified condition.

```
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

WHERE

- Select all column from **orderdetails** table where quantityOrdered greater than 75.

```
SELECT
    *
FROM
    orderdetails
WHERE
        quantityOrdered > 75;
```

orderNumber	productCode	quantityOrdered	priceEach	orderLineNumber
10401	S700_2466	85	98.72	10
10401	S700_3167	77	73.60	9
10404	S12_3990	77	67.03	4
10404	S18_3278	90	67.54	6
10405	S12_4675	97	115.16	5
10405	S24_3856	76	127.79	3
10407	S18_1749	76	141.10	2
10407	S24_2766	76	81.78	6

WHERE

- Select all column from **customers** table where **customerName** is started with ‘A’ character.

```
SELECT
    customerName
FROM
    customers
WHERE
    customerName LIKE 'A%';
```

customerName
Atelier graphique
Australian Collectors, Co.
American Souvenirs Inc
AV Stores, Co.
Auto-Moto Classics Inc.
Asian Shopping Network, Co
ANG Resellers
Alpha Cognac
Amica Models & Co.
Auto Associés & Cie.
Anna's Decorations, Ltd
Australian Gift Network, Co
Asian Treasures, Inc.
Auto Canal+ Petit
Anton Designs, Ltd.
Australian Collectables, Ltd

ORDER BY

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ORDER BY

- The ORDER BY keyword is used to sort the result-set in ascending or descending order.
- The ORDER BY keyword sorts the records in ascending order by default. To sort the records in descending order, use the DESC keyword.

```
SELECT column1, column2, ...
FROM table_name
ORDER BY column1, column2, ... ASC/DESC;
```

ORDER BY

- Show all columns from payments tables and data will be ordered by lowest amount.

```
SELECT
    *
FROM
    payments
ORDER BY amount;
```

customerNumber	checkNumber	paymentDate	amount
398	JPMR4544	2005-05-18	615.45
381	MS154481	2003-08-22	1128.20
121	FD317790	2003-10-28	1491.38
381	CC475233	2003-04-19	1627.56
103	OM314933	2004-12-18	1676.14
456	MO743231	2004-04-30	1679.92
350	OB648482	2005-01-29	1834.56
172	AD832091	2004-09-09	1960.80
161	BR352384	2004-11-14	2434.25
148	DD635282	2004-08-11	2611.84
323	HG738664	2003-07-05	2880.00
216	BG407567	2003-05-09	3101.40
219	BN17870	2005-03-02	3452.75
484	GK294076	2004-10-26	3474.66
148	ME497970	2005-03-27	3516.04
205	GL756480	2003-12-04	3879.96
204	IS150005	2004-09-24	4424.40
219	BR941480	2003-10-18	4465.85
209	ED520529	2004-06-21	4632.31
145	CN328545	2004-07-03	4710.73
181	GQ132144	2003-01-30	5494.78
256	EP227123	2004-02-10	5759.42
198	IS946883	2004-09-21	5858.56

ORDER BY

- Show all columns from payments tables and data will be ordered by highest amount.

```
SELECT
    *
FROM
    payments
ORDER BY amount DESC;
```

customerNumber	checkNumber	paymentDate	amount
141	JE105477	2005-03-18	120166.58
141	ID10962	2004-12-31	116208.40
124	KI131716	2003-08-15	111654.40
148	KM172879	2003-12-26	105743.00
124	AE215433	2005-03-05	101244.59
321	DJ15149	2003-11-03	85559.12
124	BG255406	2004-08-28	85410.87
167	GN228846	2003-12-03	85024.46
124	ET64396	2005-04-16	83598.04
114	MA765515	2004-12-15	82261.22
239	NQ865547	2004-03-15	80375.24
323	AL493079	2005-05-23	75020.13
141	IN446258	2005-03-25	65071.26
141	KT52578	2003-12-09	63843.55
157	NN711988	2004-09-07	63357.13
298	LF501133	2004-09-18	61402.00
259	EU280955	2004-11-06	61234.67
141	HJ32686	2004-01-30	59830.55
450	EF485824	2004-06-21	59551.38
495	BH167026	2003-12-26	59265.14
151	IP568906	2003-06-18	58841.35
151	BF686658	2003-12-22	58793.53
458	OO606861	2003-06-13	57131.92

ORDER BY

- Show three columns (productCode, productName, buyPrice) from products tables and data will be ordered by top cheapest items.

```
SELECT
   productCode, productName, buyPrice
FROM
   products
ORDER BY buyPrice;
```

productCode	productName	buyPrice
S24_2840	1958 Chevy Corvette Limited Edition	15.91
S24_2972	1982 Lamborghini Diablo	16.24
S24_2022	1938 Cadillac V-16 Presidential Limousine	20.61
S24_3969	1936 Mercedes Benz 500k Roadster	21.75
S24_1937	1939 Chevrolet Deluxe Coupe	22.57
S18_4668	1939 Cadillac Limousine	23.14
S32_2206	1982 Ducati 996 R	24.14
S18_2625	1936 Harley Davidson El Knucklehead	24.23
S18_1367	1936 Mercedes-Benz 500K Special Roadster	24.26
S18_2432	1926 Ford Fire Engine	24.92
S32_2509	1954 Greyhound Scenicruiser	25.98
S24_3420	1937 Horch 930V Limousine	26.30
S32_3207	1950's Chicago Surface Lines Streetcar	26.72
S50_1341	1930 Buick Marquette Phaeton	27.06
S24_1628	1966 Shelby Cobra 427 S/C	29.18
S24_3949	Corsair F4U (Bird Cage)	29.34
S12_3990	1970 Plymouth Hemi Cuda	31.92
S24_4620	1961 Chevrolet Impala	32.33
S24_1444	1970 Dodge Coronet	32.37
S72_1253	Boeing X-32A JSF	32.77

ORDER BY

- Show three columns (productCode, productName, buyPrice) from products tables and data will be shown only top 5 most expensive items.

```
SELECT
   productCode, productName, buyPrice
FROM
   products
ORDER BY buyPrice DESC
LIMIT 5;
```

productCode	productName	buyPrice
S10_4962	1962 LanciaA Delta 16V	103.42
S18_2238	1998 Chrysler Plymouth Prowler	101.51
S10_1949	1952 Alpine Renault 1300	98.58
S24_3856	1956 Porsche 356A Coupe	98.30
S12_1108	2001 Ferrari Enzo	95.59

Thank You!

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